

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

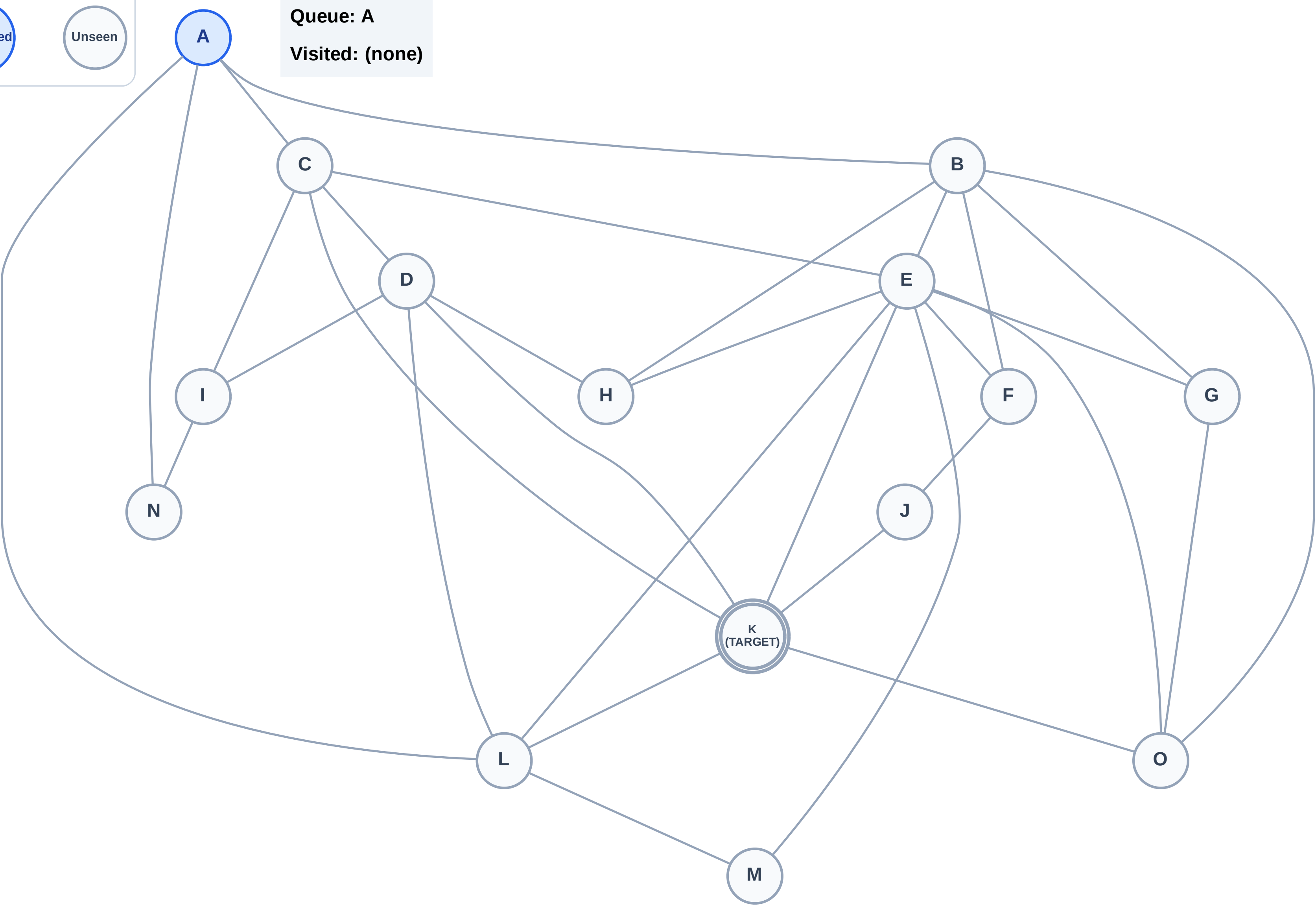
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

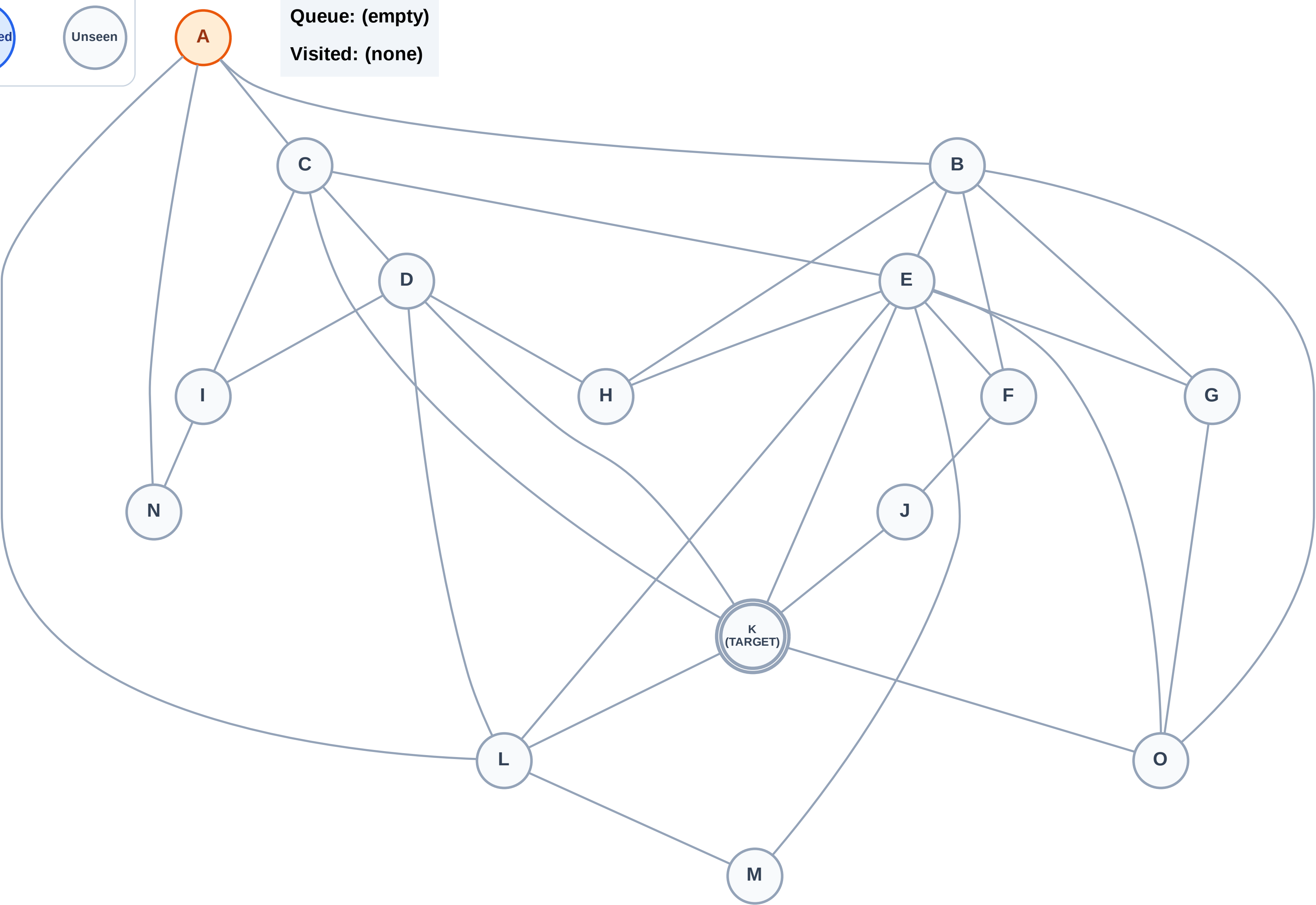
Complete

Processing

Discovered

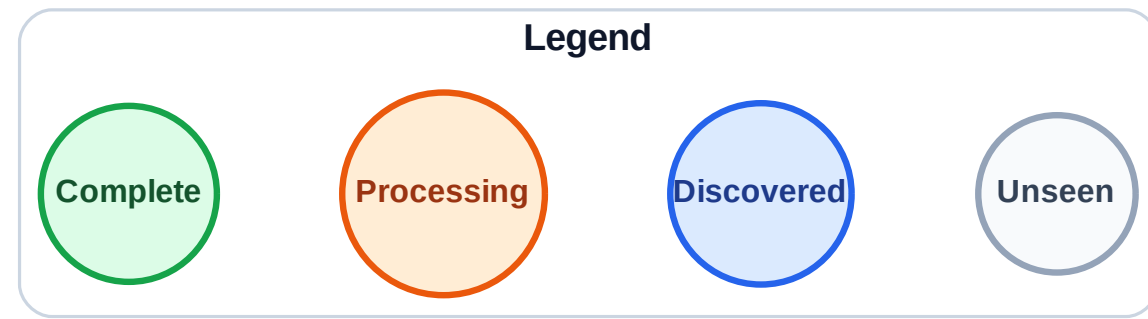
Unseen

Queue: (empty)
Visited: (none)

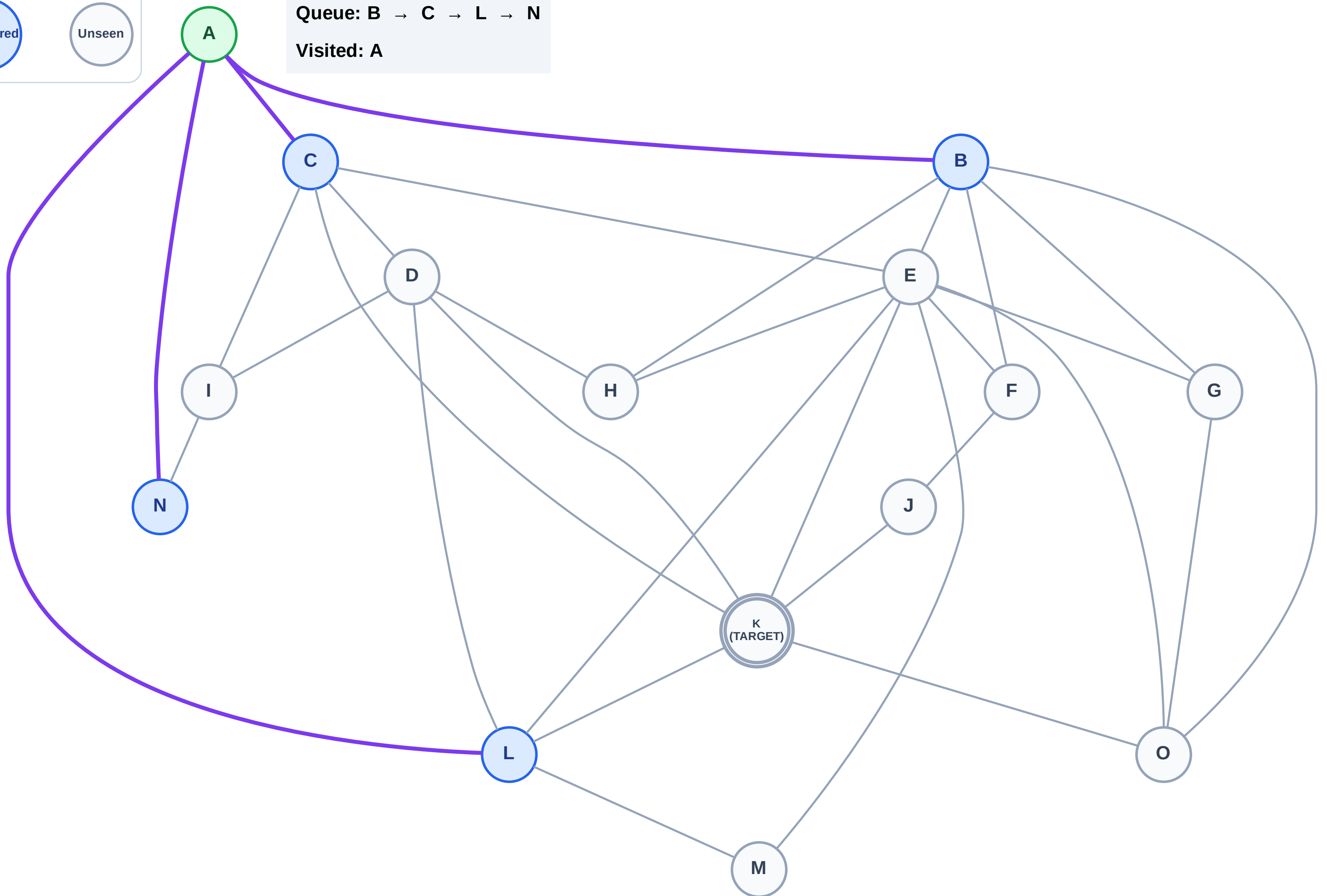


BFS Traversal

Step 3: Discover B, C, L, N from A



Queue: B → C → L → N
Visited: A



BFS Traversal

Step 4: Process node B

Legend

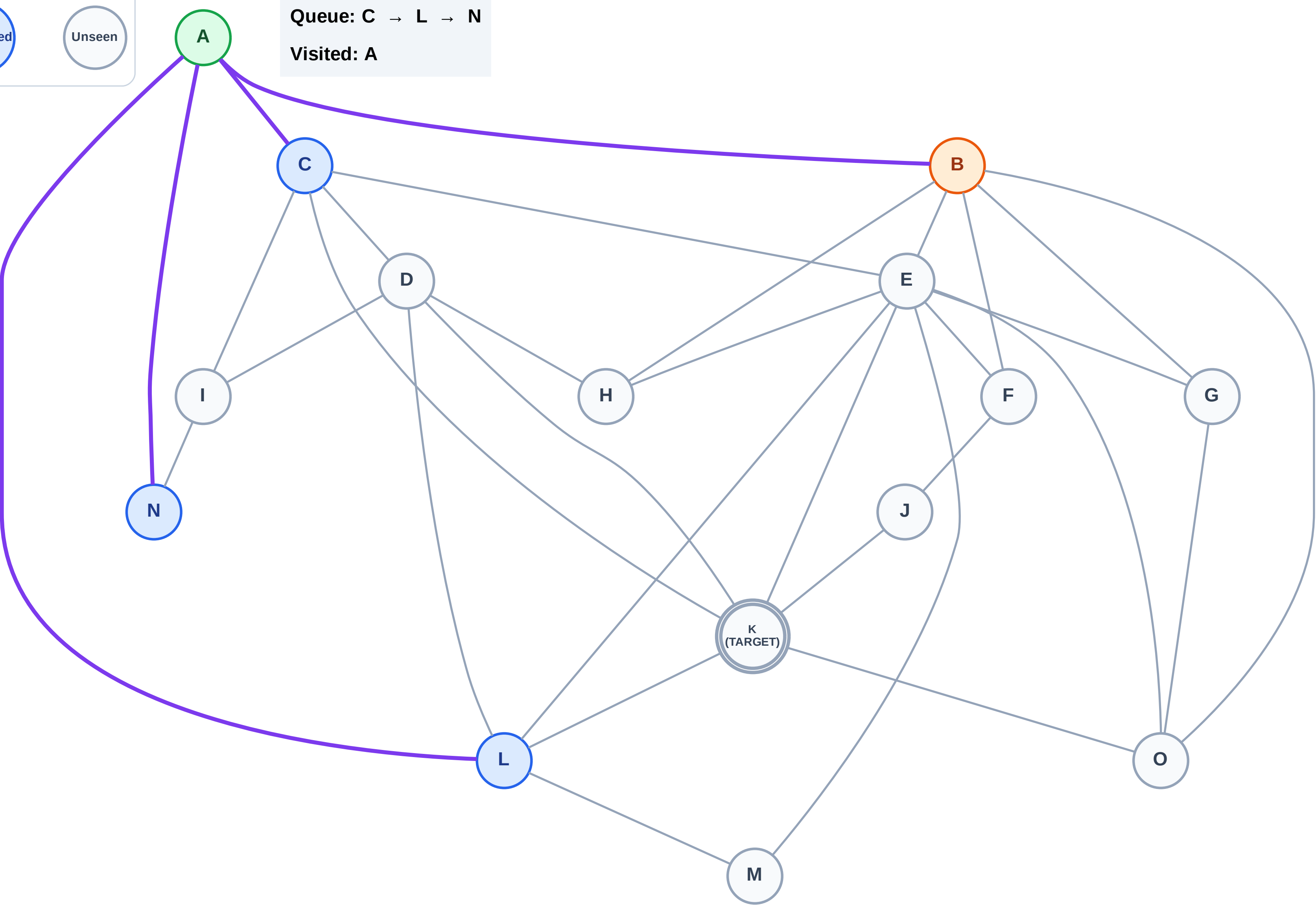
Complete

Processing

Discovered

Unseen

Queue: C → L → N
Visited: A



BFS Traversal

Step 5: Discover E, F, G, H, O from B

Legend

Complete

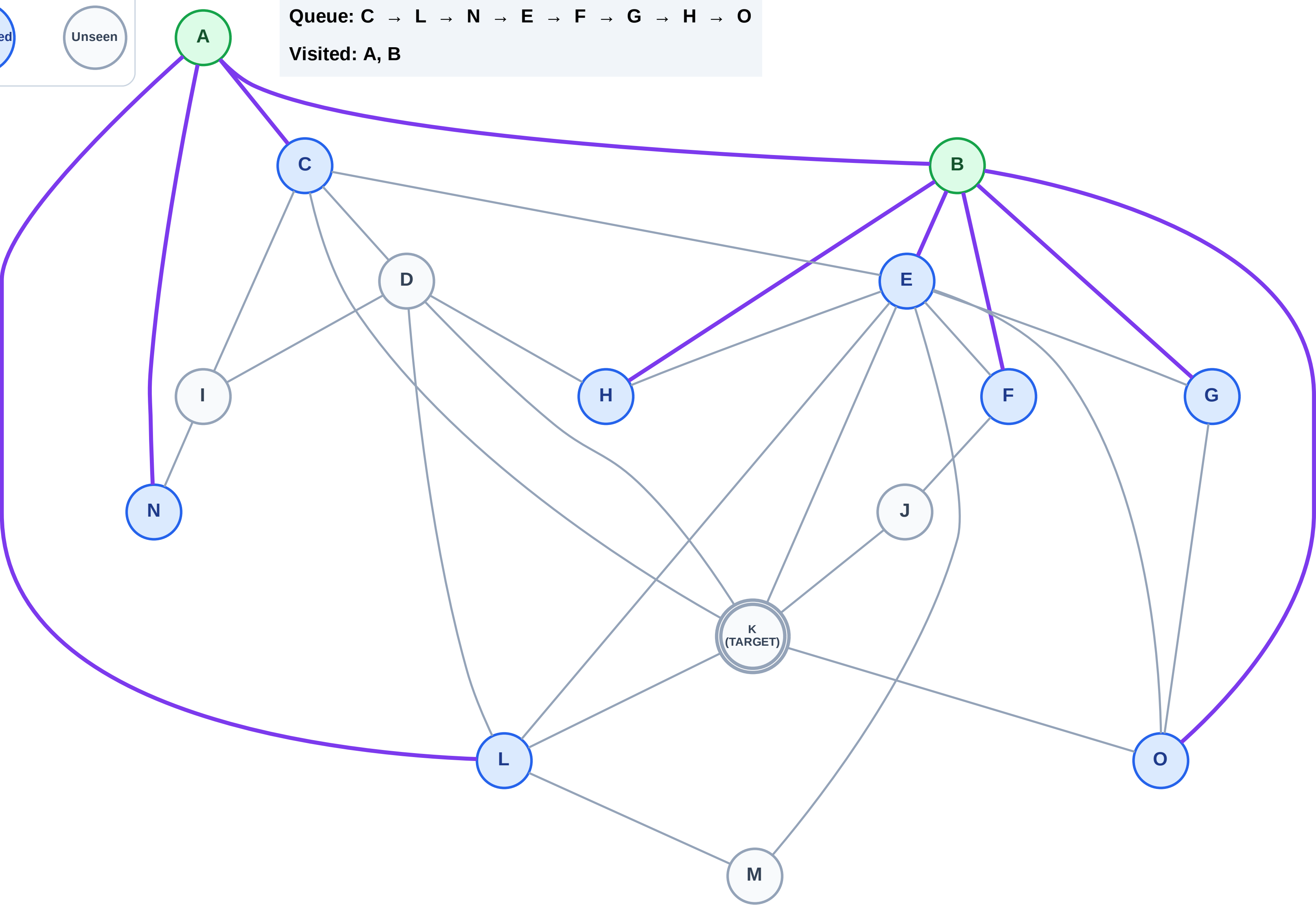
Processing

Discovered

Unseen

Queue: C → L → N → E → F → G → H → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

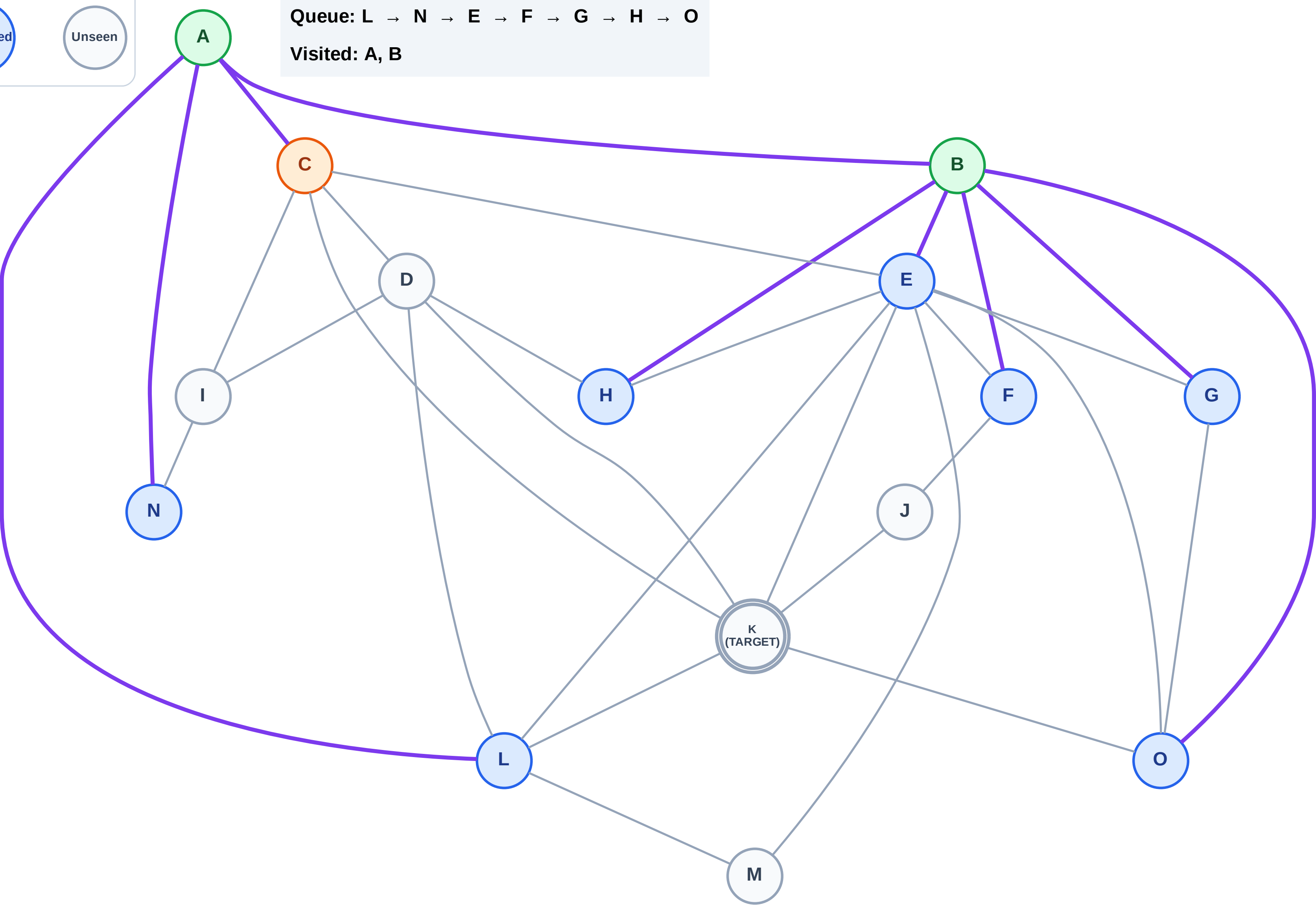
Processing

Discovered

Unseen

Queue: L → N → E → F → G → H → O

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

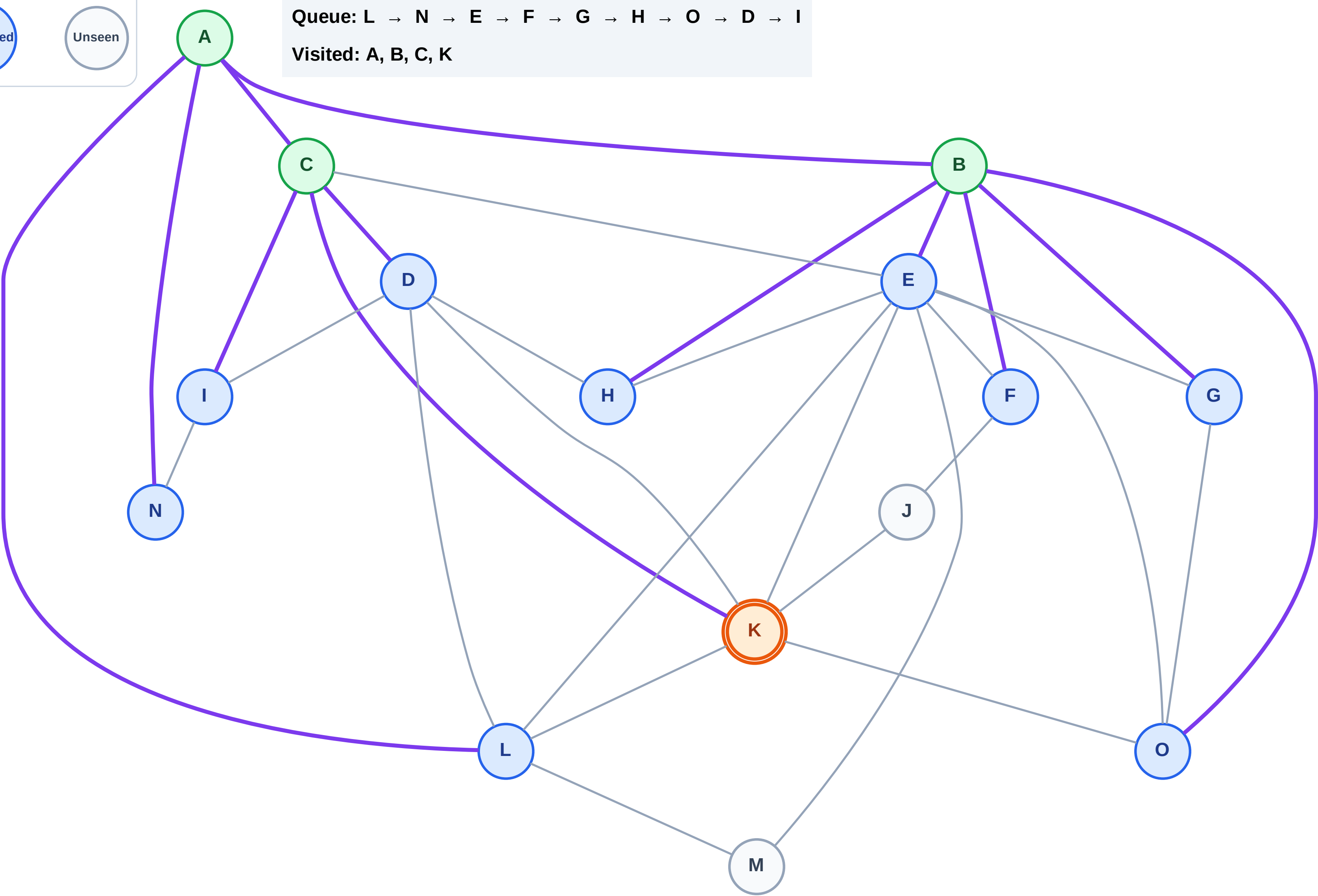
Processing

Discovered

Unseen

Queue: L → N → E → F → G → H → O → D → I

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

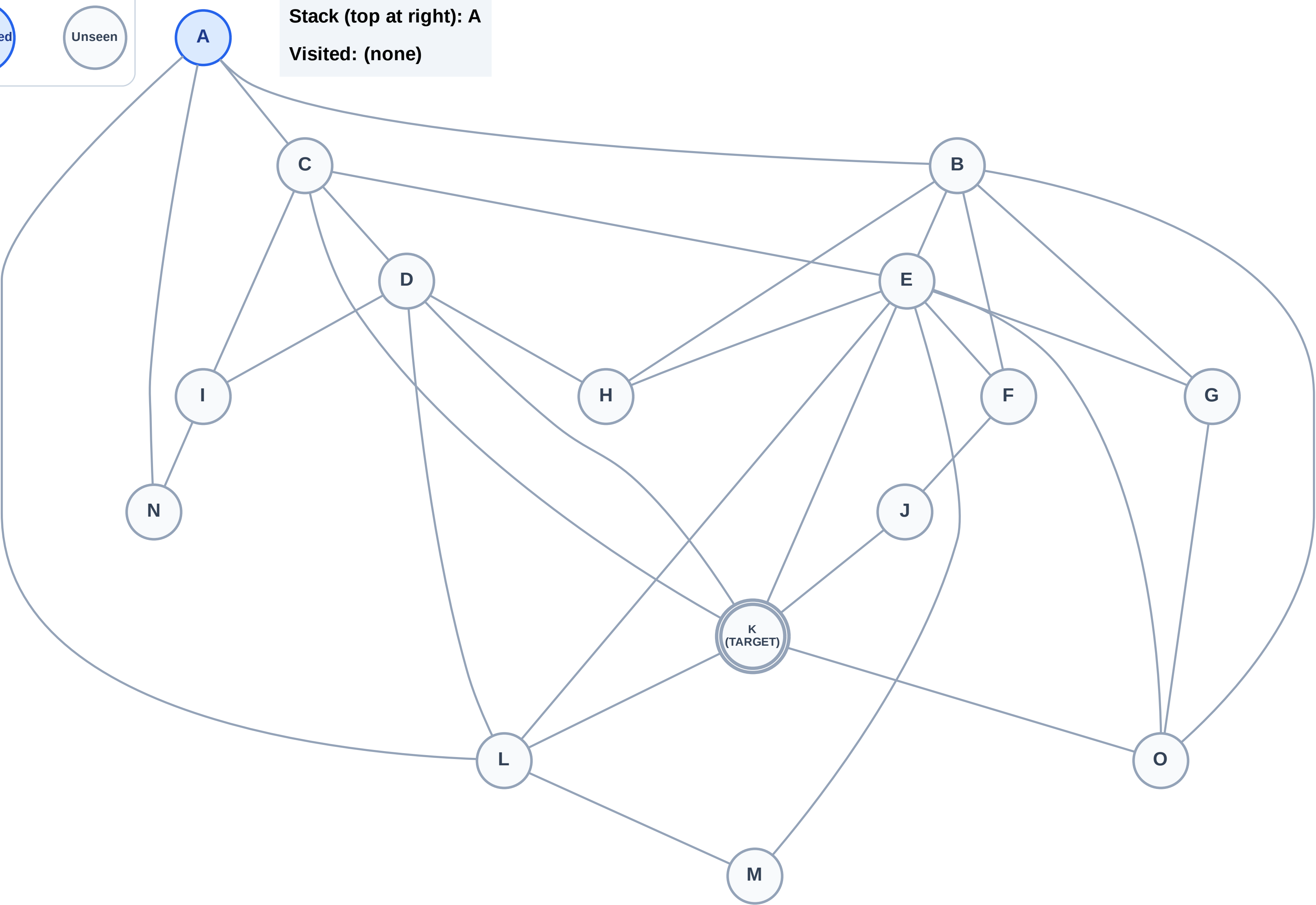
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

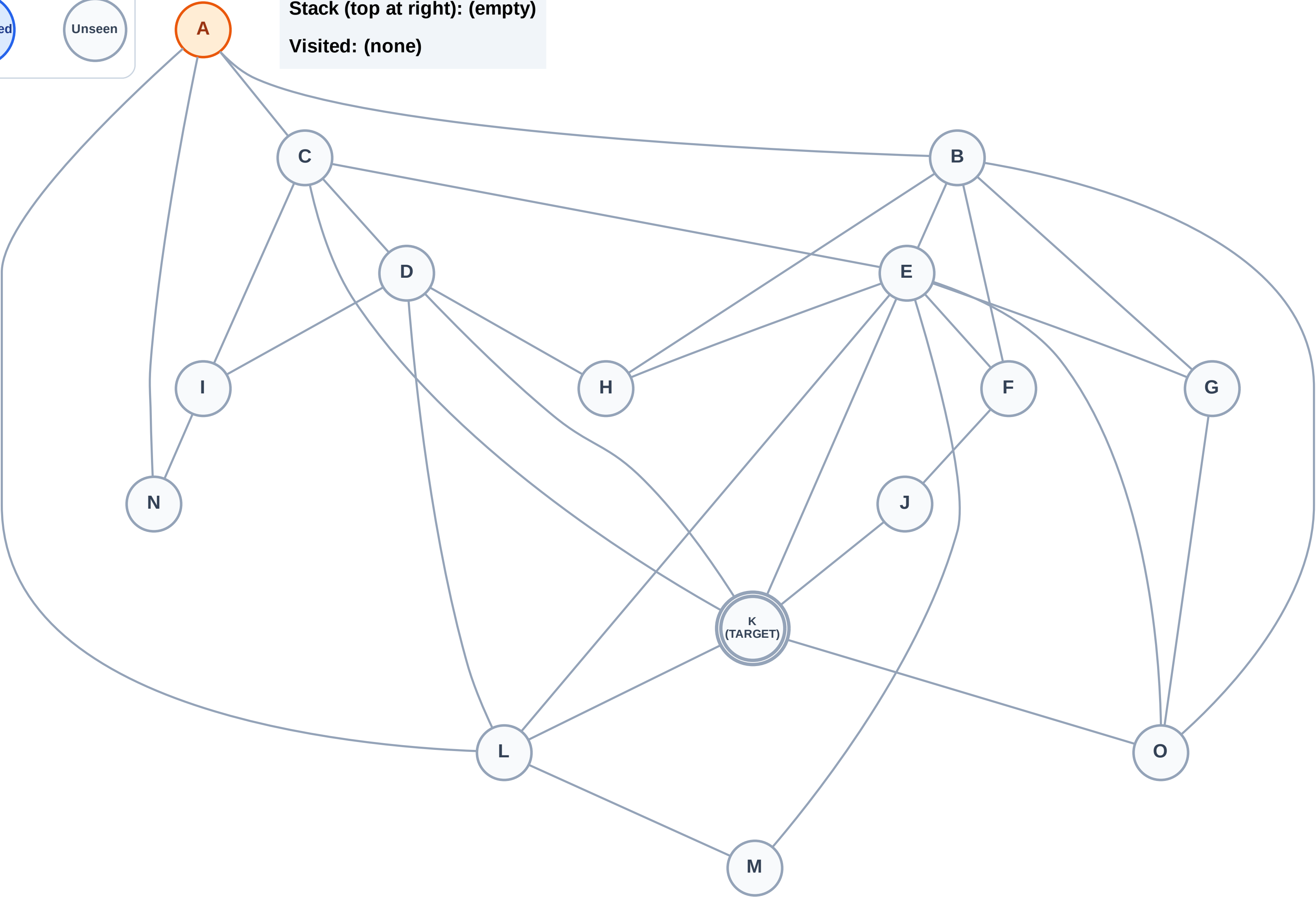
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover N, L, C, B from A

Legend

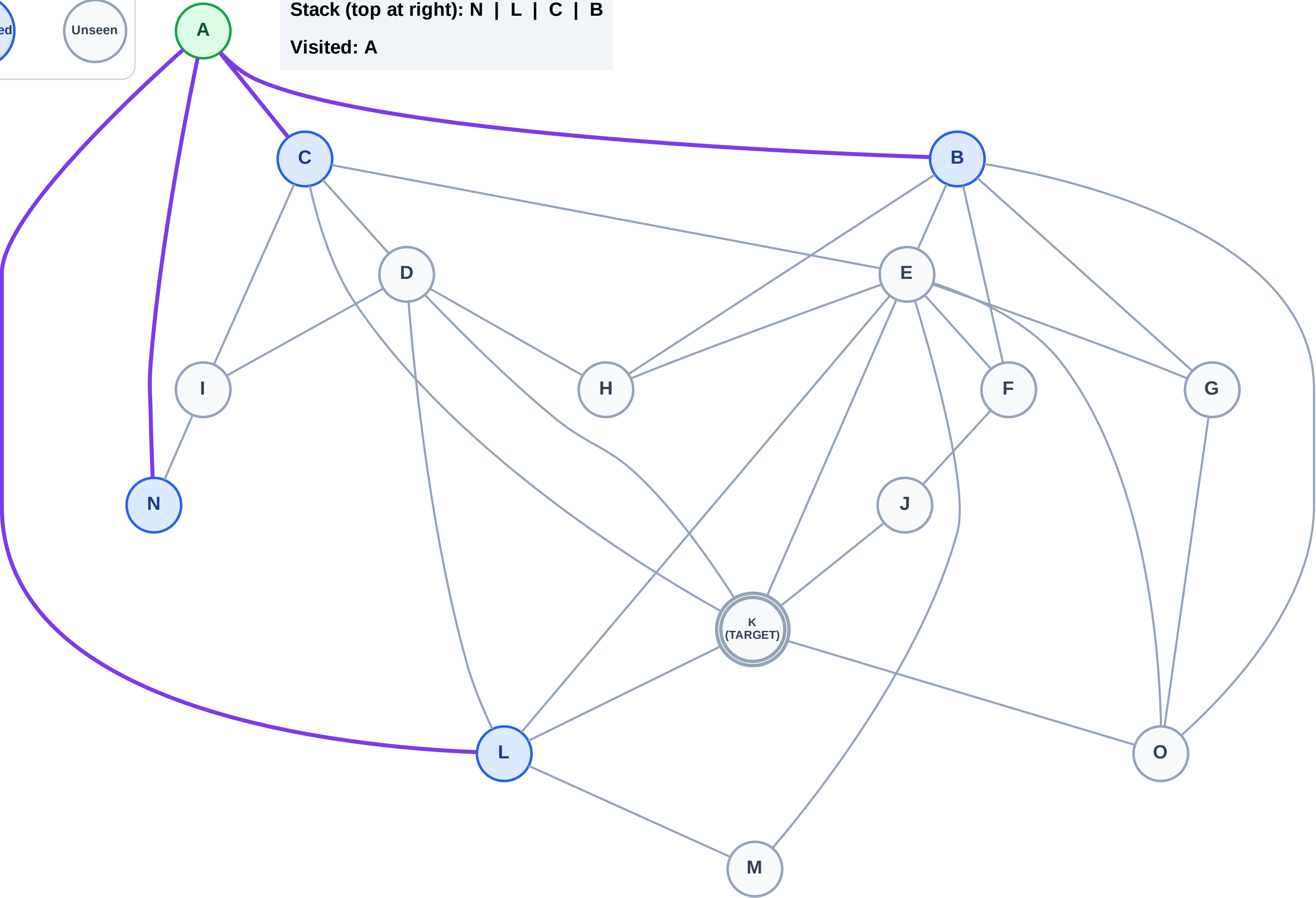
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | B
Visited: A



DFS Traversal

Step 4: Process node B

Legend

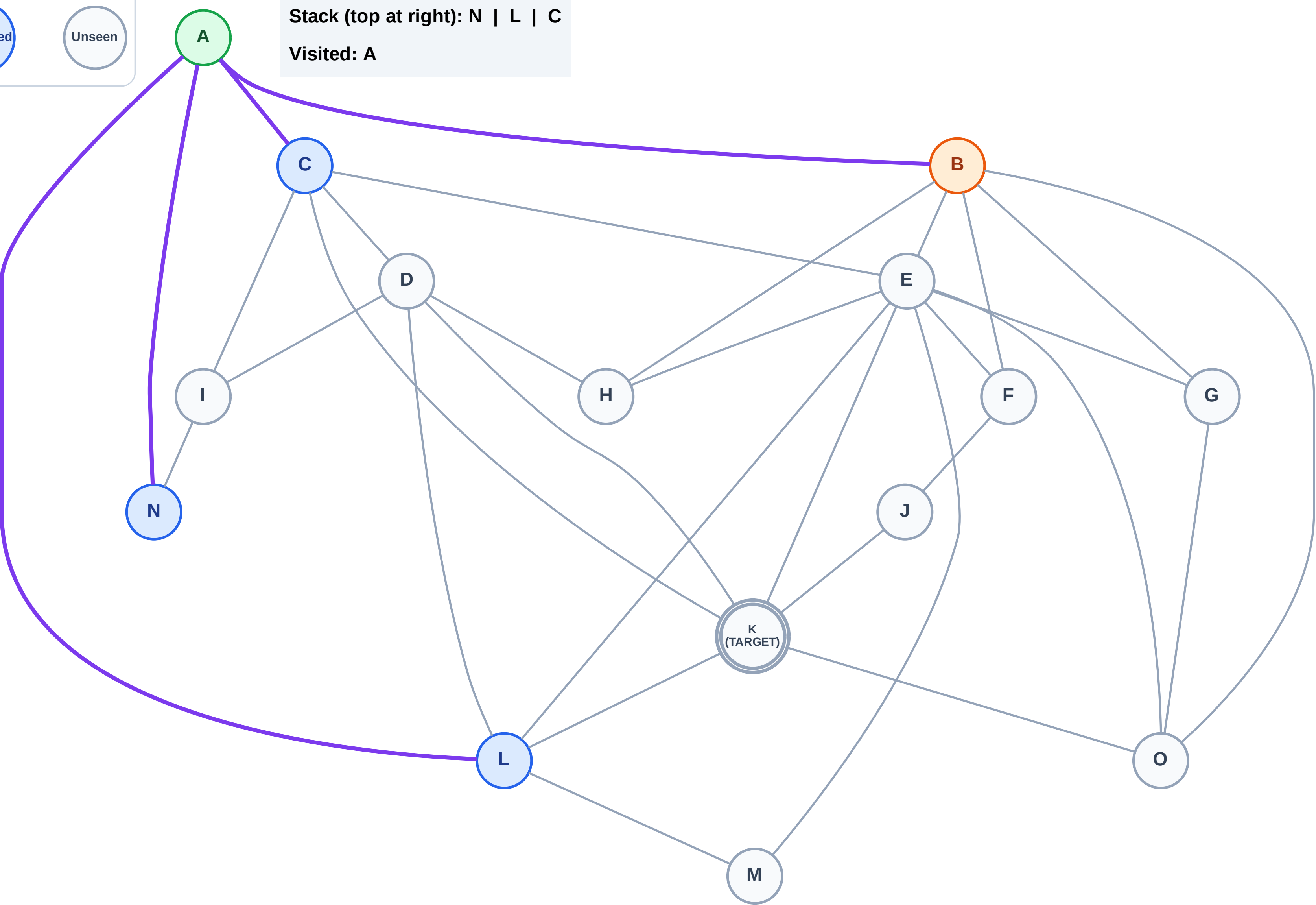
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C
Visited: A



DFS Traversal

Step 5: Discover O, H, G, F, E from B

Legend

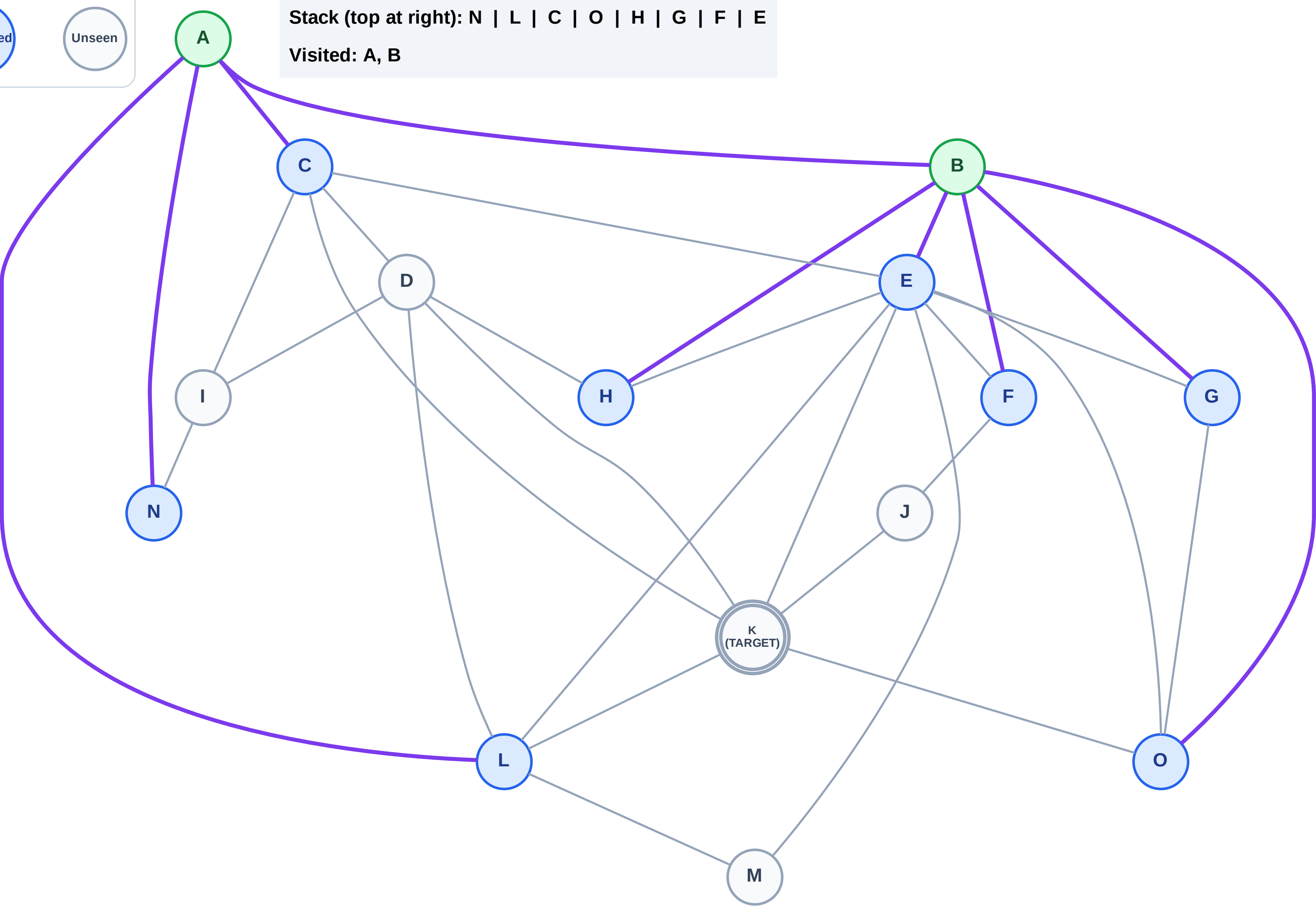
Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F | E
Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

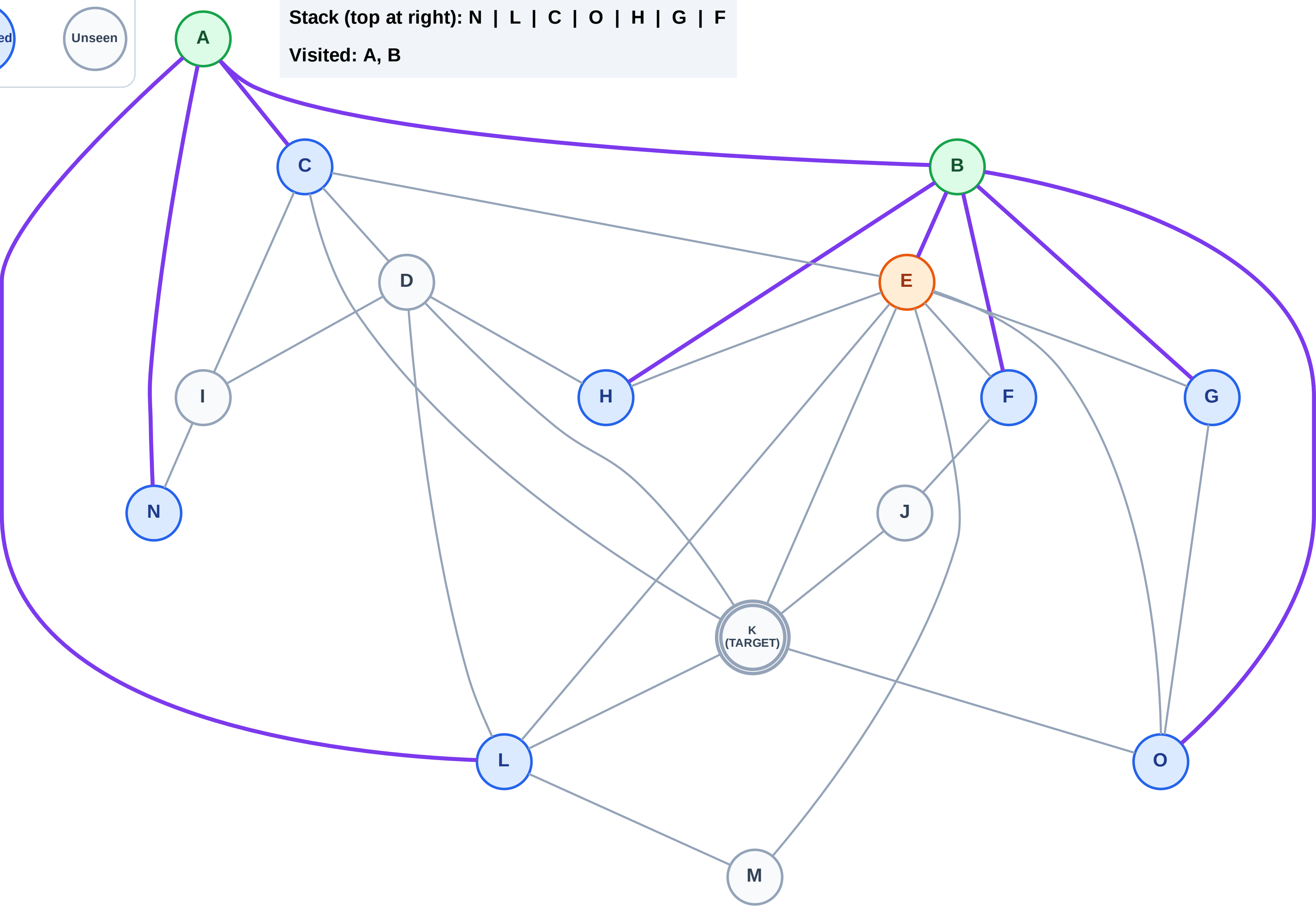
Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | C | O | H | G | F | M

Visited: A, B, E, K

